Assessment: Technical Product Manager

Candidate Name: Pranjal Jaykumar Singh

Roll No: 23-ITC27-27

Submission Document: 23-ITC27-27 Pranjal Jaykumar Singh TPM.pdf

Target Role: Technical Product Manager (TPM)

Live Deployed App (Vercel): quarantinecare-live.vercel.app

GitHub Repository: github.com/prancodes/Factwise-Assessment-TPM

Problem Statement: You are tasked with building an app for a virus quarantine & treatment facility From the head doctor:

> "This virus has a 15% mortality rate, and its main symptom is a prolonged fever. A patient is considered cured once they have reported 3 full days without a fever. Currently, our nurses jot down temperatures in a journal. The doctor reviews these journals when deciding what treatment to administer and informs the admin staff when they feel a patient could be discharged (solely based on the 3-day no-fever criterion).
>
> We are understaffed and unable to keep up with the volume of patients coming in. We have rooms for up to 74 patients, which are currently at full occupancy.
>
> Our process requires that each patient have their temperature measured by a nurse daily, and then be visited by a doctor at some point on the same day.
>
> However, in reality, no one knows which patients have had their temperatures measured, and which patients have been visited by a doctor already. So, sometimes a patient's temperature is measured multiple times on the same day, and sometimes their temperature is not measured at all. Doctor visits are also similarly erratic, and sometimes the doctor visits a patient before their temperature has been taken – as a consequence, the doctor has to waste precious minutes on doing this task themself or come back later. At times, patients end up staying an extra day or two in our centre because of these inefficiencies, or because doctors fail to promptly inform admin staff about discharges.
>
> We want to treat as many patients as possible, so we need to be better organized, and discharge patients in a timely manner. We also want to track the quality of our treatment, so we can take measures in case the mortality rate in our centre rises above the expected 15%. Currently, we have no way of measuring our performance against the 85% success rate benchmark."

You and a small team of developers need to build an app to help this centre with its temperature tracking and administration needs. Your team needs to deliver a working product ASAP to solve the centre's basic requirements, and nice-to-have features can be prioritized later.

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Q1) In your app's first version, what are the different kinds of users, and which features and actions are relevant for which user types?

To resolve the facility's bottlenecks and paper journal blindness, the v1.0 architecture implements role-based access control with three operational personas and dedicated workflows:

1. Nurse (Daily Diagnostic & Temperature Intake)

- **Pending Morning Triage Queue:** Dynamically filters the 74 beds to display only patients awaiting their daily temperature check.
- **Rapid Temperature Logger:** Single-tap presets (98.4°F, 98.6°F, 99.8°F, 101.5°F) or custom modal with °F/°C toggle and validation (93.0°F to 108.0°F).
- **Duplicate Entry Prevention:** Intercepts re-measurement attempts on the same calendar date with an explicit confirmation dialog.
- **Emergency Escalation Flag:** Immediate alert for hyperpyrexia (≥ 103.0°F) to prioritize urgent physician intervention.

2. Doctor (Clinical Evaluation & Discharge Recommendation)

- **Sequential "Ready for Round" Queue:** Strictly displays patients whose morning temperature has already been logged by a nurse, preventing wasted visits.
- **3-Day Fever-Free Streak Tracker:** Real-time clinical counter auditing consecutive days without fever (< 99.5°F).
- **Discharge Recommendation Authorization:** Action button enabled strictly when the 3-day no-fever milestone is met; automatically dispatches an instant notification to Admin.
- **Clinical Regimen & Mortality Log:** Enter treatment notes, update medication plans, and record deceased outcomes.

3. Administrator / Facility Manager (Bed Turnover & Quality Benchmarking)

- **74-Bed Facility Matrix:** Bird's-eye view of all 74 rooms color-coded by status (Awaiting Nurse, Ready for Doctor, Cured, Vacant).
- **Immediate Discharge Desk:** 1-click execution for doctor-authorized discharges, immediately freeing the room and preventing the 1-2 day bed lag.
- **Hard 74-Bed Capacity Guard:** Strict validation blocking new patient admissions when occupancy reaches 74/74.
- **Quality Benchmark Monitor:** Real-time tracking of mortality rate vs the 15% benchmark and recovery rate vs 85% target.

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Q2) Functional Prototype (Vercel Deployment & Executable Workflows)

🌐 **Live Deployed Application Link:** https://quarantinecare-live.vercel.app

📂 Open Source GitHub Repository: https://github.com/prancodes/Factwise-Assessment-TPM

The web application is built with React 19 + Vite and features full client-side state simulation demonstrating all required screens and clinical workflows:

Core Screens & UI Architecture

- **Executive Facility Dashboard:** Real-time KPI widgets tracking 74 beds occupancy, morning nurse compliance %, doctor round compliance %, and mortality vs 15% benchmark.
- **74-Room Directory:** Live search by room #/name and filtering by clinical state (*Awaiting Nurse*, *Ready for Doctor*, *3-Day Cured / Discharge Ready*, *High Fever*).
- **Nurse Triage Station:** Rapid temperature logging workflow with instant streak calculation.
- **Doctor Rounds Station:** Sequential round verification with locked discharge controls until streak == 3 days.
- **Admin Bed Turnover Desk:** One-click formal discharge releasing beds for immediate re-intake.
- **Patient Clinical Dossier:** Full temperature chart, medications list, and physician notes.
- **Interactive Assessment Modal:** Full deliverables viewer with Q1-Q4 answers and Knapsack optimizer built into the UI.

Interactive Multi-Day Simulation Engine

- **"Simulate Next Day (+1)" Button:** Advances facility calendar day, archives previous vitals, and tests the 3-day fever-free streak progression in real-time.
- **Scenario Reset:** Restores the default 74-patient occupancy state for repeated testing.

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Q3) For the entire app overview, what additional logical constraints would you provide to the developers?

The following critical business logic, state machines, and validation rules must be enforced across frontend and backend:

1. Sequential Clinical Workflow Dependency (Nurse → Doctor)

- A patient cannot be marked as "Visited by Doctor" until today's morning temperature is recorded by a nurse.
- Doctor visits before temperature logging trigger an explicit confirmation warning: *"Temperature not recorded today. Proceed anyway with manual entry?"* to prevent wasted physician rounds.

2. Strict 3-Day Fever-Free Reset State Machine

- Fever threshold is defined as ≥ 99.5°F (37.5°C). Normal body temperature is < 99.5°F.
- If a temperature ≥ 99.5°F is recorded on Day 1, 2, or 3, the consecutive fever-free counter immediately drops to 0.
- The "Recommend Discharge" button is strictly locked until `feverFreeDays == 3`.

3. Duplicate Measurement Prevention

- If a nurse attempts to log a temperature for a patient who already has a reading for that calendar date, direct overwrite is blocked.
- The user must confirm via modal: *"Temperature already recorded today (XX.X°F at HH:MM). Confirm if you are logging a re-check or correcting an error."*

4. Hard 74-Bed Ceiling & Sanitization Cycle

- Concurrent active patients cannot exceed 74. When occupancy is 74/74, the "New Admission" button is disabled.
- Discharged beds move to a `Cleaning / Sanitizing` state before becoming available for new patient intake.

5. Quality Metric Calculation Rule

- Mortality rate is computed strictly over closed/resolved cases: `Mortality Rate = [Deceased / (Discharged + Deceased)] * 100%`.
- Active patients in quarantine are excluded from the denominator to avoid skewing early metrics. An automated alert fires if mortality > 15.0%.

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Q4) Sprint Planning & Knapsack Optimization Analysis

Problem Parameters: Velocity = 25 story points | Already planned = 3 stories × 5 points = 15 points | **Remaining Capacity = 10 story points**.
Value Weighting: Low = **1 unit** | Medium = 3× Low = **3 units** | High = 2× Medium = **6 units**.

Feature	Priority	Effort (pts)	Value	Value Density (ROI)	Decision & Rationale
A	High	1.0	6	**6.00**	**SELECT:** Exceptional ROI, highest efficiency.
B	High	3.0	6	**2.00**	**SELECT:** High strategic value, strong density.
C	High	5.0	6	1.20	**EXCLUDE:** Consumes 50% capacity for low ROI (1.20).
D	Medium	2.0	3	**1.50**	**SELECT:** High-density medium priority feature.
E	Medium	3.0	3	1.00	**EXCLUDE:** Dominated by Feature D (same value, 1 more effort).
F	Medium	1.0	3	**3.00**	**SELECT:** Outstanding medium priority ROI (3.00).
G	Low	0.4	1	**2.50**	**SELECT:** High density quick-win.
H	Low	0.5	1	**2.00**	**SELECT:** High density quick-win.
I	Low	0.3	1	**3.33**	**SELECT:** Ultra-high density quick-win.
J	Low	0.1	1	**10.00**	**SELECT:** Highest density feature in backlog (10.00).

Final Mathematical Recommendation for Sprint Backlog:
Selected Feature Set: `{ A, B, D, F, G, H, I, J }`
- **Total Effort Expended:** 1.0 + 3.0 + 2.0 + 1.0 + 0.4 + 0.5 + 0.3 + 0.1 = **8.3 story points** (≤ 10 capacity).
- **Total Delivered Value:** 6 + 6 + 3 + 3 + 1 + 1 + 1 + 1 = **22 value points** (global theoretical maximum).
- **Sprint Risk Buffer: 1.7 story points** left as contingency buffer against velocity variance.

Why this dominates naive sprint planning:
A naive approach would pick all High features `{A, B, C, F}`, consuming all 10 points for only 21 value. By replacing low-ROI Feature C (effort 5.0, value 6) with high-density features `{D, F, G, H, I, J}` (effort 4.3, value 10), we deliver **more value (+1 point / 22 total)** with **less effort (–1.7 points / 8.3 total)**.

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